EXP. NO. 1

23/07/24 Study of Various Network commands used in Linux and Windows

Aim: To Study various network commands in linux and windows

(1) arp -a : ARP is short form of address resolution Protocol. It will show the IP address of your computer along with Ip address, MAC address of your router

O/P: gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3.

(2) hostname : This is the simplest of all TCP/IP commands It simply displays the name of your computer

O/P: localhost. localdomain

(3) ipconfig/all : This commands display detailed configuration about your TCP/IP connection including router, gateway DNS, DHCP, type of Ethernet adapter in your system.

O/P: enp0s3 : flags=4163<UP, BROADCAST, RUNNING, MULTICAST> mtu 1500, inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255

lo : flags = 73 <UP, LOOPBACK, RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet 6 ::1 prefixlen 128 scopeid 0x10<host>

(4) nbstat -a : This command helps solve problems with NetBios name resolution (Nbt stands for NetBios over TCP/IP)

O/P: Ethernet
Node IpAddress : [0.0.0.0] Scope Id : []
Bluetooth network connection :
Node Ip address : [0.0.0.0] scope Id : []

(5) netstat -a : (network statistics) network displays a variety of statistics about computer active TCP/IP connection. It is a command line tool for monitoring network connection both incoming, outgoing as well as routing tables, interface statistics etc.

O/P : Kernel IP routing table

Destination	Gateway	Genmask	Flag	MSS	Window	irtt	Iface
default	-gateway	0.0.0.0	UG	0	0	0	en
10.0.2.0	0.0.0.0	255.255.255.0	U	0	0	0	en

(6) nslookup : (name server lookup) is a tool used to perform DNS lookup in linux. It is used to display DNS details such as IP address of a particular computer. nslookup can operate in two modes interactive, non interactive

O/P : Server : 10.86.248.99
address : 10.26.248.99.99 #53

(7) pathping : pathping is unique to window and is basically combination of ping and tracert commands. It traces the route to destination address then launches 25 second test of each router.

O/P: Tracing route to google.com [2404:6800:4013:
800::8b]
over maximum of 30 hops :

0. Desktop-TV7UP9D

1. 2409:4072:2ec2:9377::82

(9) ping : (Packet Internet Groper) command is the best way to test connectivity between two nodes. ping via ICMP to communicate to other devices.

1. # ping hostname (ping localhost)
2. # ping ip address (ping 4.2.2.2)
3. # ping fully qualified domain-name (ping www.facebook.com)

O/P: pinging googe.com [2404:6800:4007:838 ::200e] with 32 bytes of data :

Reply from 2404:6800:4007:838 ::200e : time =26 ms

(10) Route : Route command is used to show the IP routing table. It is primarily used to setup static routes to specific routes host or networks via an interface

O/P: Interface List.

17 ... 00 00 27 00 00 11 .... Virtualbox Host-only Ethernet Adapter

13 ..... 5a cd ca 38 02 25 ... Microsoft wifi direct Virtual Adapter # 3.

IP V4 Route table

Active routes :

Network destination	netmask	gateway	interface	metric
0.0.0.0	0.0.0.0	172.16.96.1	172.16. 102.158	30

Some important linux networking command

1. ip - can show address information, manipulate routing plus display network, various, devices interfaces, tunnels

(a) ip address show

O/P
1: lo: <LOOPBACK, LOWER_UP> mtu 65536
qdisc noqueue state unknown group default q len
1000 link/loopback 00:00:00:00:00:00
brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo valid_lft
forever preferred_lft
inet6 ::1/128 scope host valid_lft forever
preferred_lft forever

(b) ip route get
Displays the route taken for IP 8.8.8.8.

O/P 8.8.8.8 via dev 172.16.8.1 dev enp0s3
src 172.16.9.200 uid 3479 cache.

(C) ip link show - Displays all the network interfaces and shows their status

O/P: 1: lo: <LOOPBACK, UP, LOWER_UP> mtu 65536 qdisc
noqueue state unknown mode DEFAULT

(d) ip route show - Displays system routing table including default gateway, network routes

O/P default via 172.16.8.1 dev enp0s3 proto static
metric 100

172.16.8.0/22 dev enp2s0 proto kernel scope link
src 172.16.9.200 metric 100

(2) ifconfig: this command is a staple in many sysadmin's tool-belt for configuring, troubleshooting networks. I has been replaced by ip command above

O/P:

enp2s0: flag = 4163<UP, BROADCAST, RUNNING, MULTICAST> mtu 1500

inet 172.16.9.200 netmask 255.255.252.0 broadcast 172.16.11.255

inet6 fe80::cb77:eb32:8167:72fe prefixlen 64 scopeid 0x20<link>

ether 50:9a:4c:37:fa:c7 txqueuelen 1000

RX packets 147995 bytes 21856810 (20.8 MiB)

TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(3) mtr - Matt's traceroute is a program with command line interface that serves as a network diagnostic troubleshooting tool. Just like traceroute, mtr command show the route from a computer to specified host. mtr provides a lot of statistics about each hop, such as response time, percentage. With mtr, you will get more information about route, be able to see problematic devices along the way. Sudden increase + packet lo

O/P: My traceroute [V0.95]

fedora (172.16.9.200) -> google.com (142.250.77.110) 2026-07-23
:44:36-040

Keys: Help Display mode Restart Statistics Order of fields qui

Host	Loss %	Snt	Last	Avg	Best	Wrst	St
1. _gateway	14.7%	200	0.4	0.5	0.2	14.5	
2. mail.totaldo.com	10.2%	246	11.1	16.7	3.5	65.5	

b) Show numeric IP address. - mtr -g google.com

O/P: fedora (172.16.9.200) -> google.com (142.250.67.46) 2026-07-23

Keys: Help Documentation Restart Statistics order of fields quit

Host	Packets Loss	Snt	Last	Pings Avg	Best	Wrst
1. 172.16.8.1	0.0%	107	0.3	0.4	4.9	0.5
2. 182.156.9.89	2.4%	127	14.2	13.0	3.2	30.

(c) Show numeric IP with hostname - mtr -b google.com

O/P:

Host	Packets Loss %	Snt	Last	Pings Avg	Bst	Wrst	Dev
1. _gateway (172.16.8.1)	0.0%	56	6.2	2.7	0.2	12.9	S
2. mail.totaldc.com (182.156.9.89)	13.8	64	26.2	21.6	25	31.2	

(d) Set the number of pings that you want to send

O/P

Host	Packets Loss%	Snt	Pings Avg	Bst	Wrst	StDev
1. localhost	0.0%	7	0.1	0.0	0.1	0

(4) tcpdump : is designed for capturing, displaying packets

O/P: [sudo] password for kavya : root

(5) ping - is a tool verifies IP level connection to another TCP/IP computer by sending Internet control message protocol ECHO request message.

$ping google.com

O/P:

PING google.com (142.250.67.46) 56 (84) bytes of data.

64 bytes from maa05s12-in-f14.1e100.net (142.250.67.46): icmp-seq=1 ttl=118 time = 2.43ms

$ ping -c 10 google.com

O/P 10 packets transmitted, 10 received, 0% packet loss, time 9016ms rtt min/avg/max/dev = 5.644/5.676/15.274/3.392ms

Result:

Thus all the networking commands were executed.